

PAPER_956: Spectral Ladder Phonon Mapping (26-Level)

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** et_phonon_resonance.py §7 (SpectralLadderPhononMapping) **Calculator:** SpectralLadderPhononMappingCalc (CP4 #540) **CVW:** v2.0.0 compliant

Abstract

We construct the phonon mapping for the 26-state HRes spectral ladder, converting each energy level $E_n = E_0 \cdot (2\pi)^{n/3} \cdot S_{26}$ to its corresponding phonon frequency ω_n and quality factor Q_n . The mapping identifies which levels fall in THz, GHz, or sub-GHz phonon regimes.

1. Phonon Frequency Mapping

$$\omega_n = \frac{E_n}{\hbar} = \frac{E_0}{\hbar} \cdot (2\pi)^{n/3} \cdot S_{26}$$

2. Quality Factor

$$Q_n = \frac{\omega_n \cdot \sqrt{E_n}}{k_B T}$$

3. Regime Classification

Level Range	ω_n	Regime
n = 1-5	Sub-THz	Acoustic phonon
n = 6-15	THz	Optical phonon
n = 16-26	Multi-THz	High-frequency phonon

4. 26-Level Table

Each level n from 1 to 26 produces a unique (E_n, ω_n, Q_n) triplet, fully determined by the UQFF spectral ladder constants.

5. Source Data

- **File:** et_phonon_resonance.py §7
 - **Session:** 214
 - **CP4 Class:** SpectralLadderPhononMappingCalc (#540)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)